

# FinTech Client: QA architecture

A concrete design for AI assistance across a release workflow with 75 offshore QA staff.

75

## offshore QA staff

One payment release.

A platform the next team can reuse.

Technical SDs and leads • Fictional fintech • Proposed architecture

# The application landscape

Payment APIs offer the first controlled test surface. Other applications follow their own readiness.

## Payment services

Java / Spring Boot  
PostgreSQL journal  
Versioned payment API

### First pilot boundary



## Web + mobile

React journey; existing Selenium and Appium suites

## Provider integration

WireMock during service tests; provider sandbox for integration

## Settlement batch

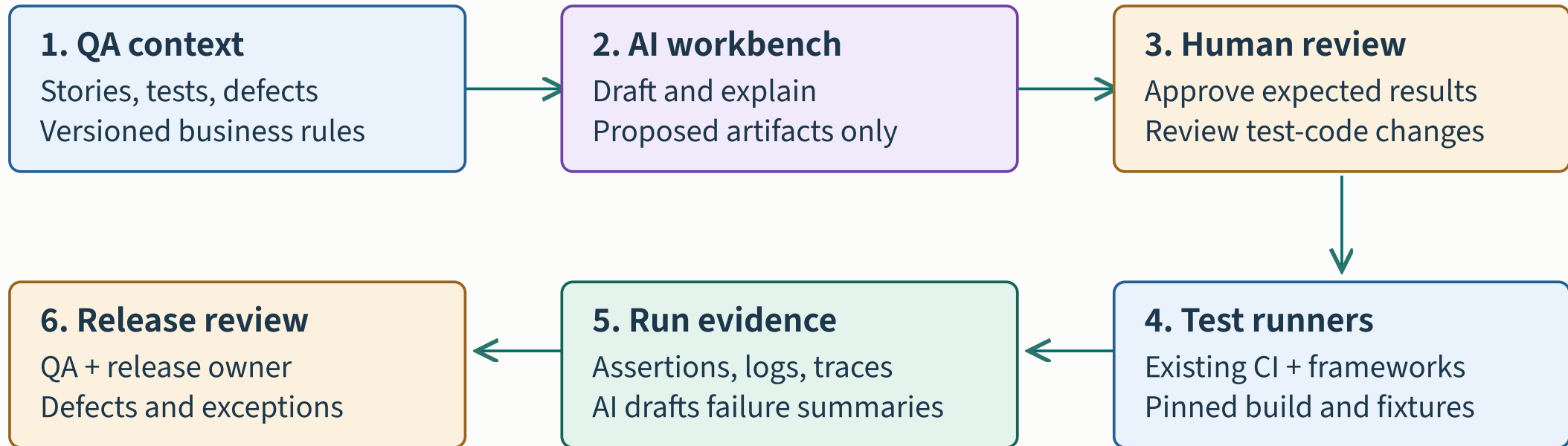
Legacy Java + SQL; shared nightly environment and file reconciliation



Example stack: Java/Spring Boot, React, PostgreSQL and Azure. Existing automation remains useful.

# The shared QA platform

The model proposes artifacts. Reviewers approve them. Existing runners produce evidence.



Shared foundation: synthetic data · dependency stubs · artifact IDs · ownership · model configuration

Adapters, context assembly and evidence linking are integration work to build. No turnkey platform is implied.

# PAY-142: the payment contract

| CONDITION                                | EXPECTED BEHAVIOR                                                               |
|------------------------------------------|---------------------------------------------------------------------------------|
| Same key + same payload                  | Return the same transfer ID and result. Post the journal once.                  |
| Same key + changed amount                | Reject the conflicting request without another posting.                         |
| Concurrent retries / duplicate callbacks | Converge on one transfer and one balanced journal.                              |
| Expected final state                     | Payer CAD 900.00; recipient CAD 100.00; one journal with two balancing entries. |

Authored business rules: initial balances CAD 1,000.00 and CAD 0.00; transfer CAD 100.00; no fees, FX or other activity. No regulatory claim.

# QE testing scenario: detecting a duplicate posting

Simulate an accepted transfer, a timed-out response and a retry. A deliberately injected defect posts the same transfer twice.

## Customer

Send CAD 100

## Payment API

→ Key: pay-142

## Provider

→ Accept; response times out

## Customer retries

→ Reuse pay-142

Injected test defect

## 2 journals

Payer CAD 800

Recipient CAD 200

**FAIL • duplicate posting**

Expected test outcome

## 1 journal

Payer CAD 900

Recipient CAD 100

**PASS • expected state**

QE testing scenario. The injected defect causes the duplicate posting; a timeout alone does not imply two debits. Initial balances: CAD 1,000 and CAD 0. One journal contains one debit and one credit. No fees, FX or other activity.

Assert one journal and correct balances. The duplicate-posting variant must fail. This is a QE test scenario, not a reported client incident.

# Requirements and test design

PAY-142 starts with an ambiguity: does a timeout mean failure, success or unknown?

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## Requirements

### AI assistance

Draft ambiguity questions and identify affected payment journeys.

### Output and owner

Versioned acceptance criteria and an open-question log. Product owner + domain QA.

### Human check

Product owner confirms that a retry uses the same key and returns the original transfer.

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## Test design

### AI assistance

Draft positive, boundary, retry and concurrency cases with requirement links.

### Output and owner

Reviewed test matrix with independently defined expected results. Domain QA lead.

### Human check

Include same-key/different-payload, parallel retry and duplicate callback cases.

Domain QA and the product owner approve expected behavior before generated tests can validate it.

# Test data and environment preparation

Synthetic accounts, known balances and a controlled provider response make the retry repeatable.

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## Test data

### AI assistance

Draft seed scripts and scenario combinations using synthetic records.

### Output and owner

Versioned fixtures with isolated accounts and known starting balances. Test-data engineer.

### Human check

Verify relationships and exact balances before running tests.

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## Environment

### AI assistance

Summarize readiness failures and draft dependency stub mappings.

### Output and owner

Reproducible test run with pinned build, data and stub versions. Platform engineer + QA.

### Human check

A deterministic preflight checks health, schema and fixture readiness.

Container-based service tests do not replace real integration, mobile or batch testing. Validate stubs against the provider contract.

# Generated code enters an ordinary review

Use the approved case matrix and repository examples to produce a candidate test change.

## 1 / AI CANDIDATE

### A weak first draft

```
assert status == 200
```

A success response can hide two journals.

## 2 / HUMAN REVIEW

### Reject and repair

Require the original transfer ID, one journal and independently agreed balances.

Unexplained skips cannot satisfy the case.

## 3 / CI CHALLENGE

### Prove the assertion

Expected balances: **90000 / 10000** minor units.

The correct build must pass; the deliberate duplicate-posting variant must fail.

Authored PAY-142 review example; pseudocode and expected results are illustrative. Other required cases and real integration checks remain necessary.

The reviewer checks the oracle. A deliberate duplicate-posting fault must fail before the candidate joins the required suite.



# The appropriate test layer

| LAYER                       | EXAMPLE IMPLEMENTATION                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| Service and API             | REST Assured + JUnit. Cover retry, concurrent requests, idempotency conflicts and journal outcomes.    |
| Web journey                 | Playwright for the new React flow. Check pending state, retry action and the final customer message.   |
| Existing browser and mobile | Keep owned Selenium and Appium coverage. Change frameworks only when maintenance evidence supports it. |
| Batch and integration       | Retain settlement-file reconciliation and provider sandbox checks with explicit scheduling.            |

Test distribution follows behavior and observability. A new browser framework does not solve a shared settlement environment.

# Execution retains required coverage

CI records the build, fixture, stub and test revisions before runners execute the pack.

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## Execution

### AI assistance

Recommend affected tests and summarize run evidence.

### Output and owner

Runner results, traces and mandatory-suite completion record. Application QA team.

### Human check

Keep the mandatory payment suite while validating selection quality.

AI test-selection recommendations run in shadow mode first. Required payment checks continue on every eligible release.

# Triage preserves the original test intent

A failed journal assertion belongs to the product defect path until evidence establishes otherwise.

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## Triage & retest

### AI assistance

Cluster related failures and draft evidence-linked defect or repair proposals.

### Output and owner

Reviewed defect disposition, fix and fresh rerun evidence. QA lead + developer.

### Human check

Separate product defects, flaky tests and environment faults. Recheck original intent after a patch.

Capture failed tests, skipped tests and retries. A repair that removes an assertion does not count as a passing outcome.

# A traceable release decision

Every generated artifact and run links back to the approved business behavior.

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## PAY-142

Approved behavior

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## Test revision

Reviewed assertions

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## Run manifest

Build, data, stub

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## Run evidence

Results + traces

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## Disposition

Defect / fix / rerun

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## Release owner

Recorded decision

AI can draft the summary. Missing assertions or unresolved defects remain visible to the release reviewer.

An absent result remains unknown. The release owner resolves exceptions and the QA lead reviews the evidence pack.

# What the demonstration proves

| TRY IN THE CASE EXPLORER | EXPECTED OUTCOME                                                         |
|--------------------------|--------------------------------------------------------------------------|
| Normal transfer          | One posting and expected balances: the modeled checks pass.              |
| Retry with injected bug  | Two postings and incorrect balances: the modeled checks block release.   |
| Retry with deduplication | One posting after two attempts: the modeled checks pass.                 |
| Duplicate callback       | Repeat the event with the same identity and inspect the journal outcome. |

The browser demonstration is a deterministic teaching model. It does not execute REST Assured, call a payment service or prove production correctness.

# Readiness across six dimensions

These dimensions describe the delivery profile in this scenario. Review the wider dependency register before approving a workflow.

| DIMENSION               | MIXED DELIVERY PROFILE                            | FIRST IMPLEMENTATION PRIORITY                      |
|-------------------------|---------------------------------------------------|----------------------------------------------------|
| Requirements            | Versioned stories; some ambiguity                 | Confirm payment outcomes before generation         |
| DevOps                  | CI available; deployment partly manual            | Pin build and deployment revisions                 |
| Cloud & environments    | Cloud test environment; data reset shared         | Isolate accounts and script data reset             |
| Application testability | Payment API testable; batch dependency shared     | Pilot the API and control provider behavior        |
| Automation              | Java automation usable; UI pack noisy             | Stabilize owned tests and retain required coverage |
| Offshore delivery       | Domain expertise strong; automation skills uneven | Pair reviews and define the next owner             |

Assess each application separately. Unknown infrastructure, data, virtualization, AI access or ownership must be resolved before the affected workflow can proceed.

# Offshore handoff and platform ownership

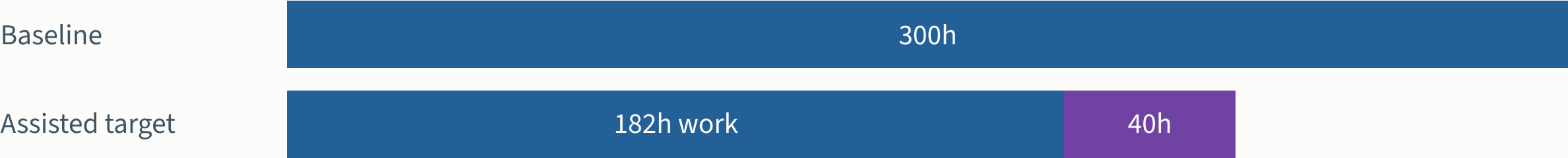
Each handoff carries a reproducible artifact and a named next owner.

| Owner / artifact        | Shared overlap                                        | Offshore workday                                            | Next overlap                                       |
|-------------------------|-------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------|
| Product + domain QA     | Approve PAY-142<br>Rule + scenario matrix ↓           | Questions wait for a named reviewer                         | Resolve ambiguity<br>Updated rule → QA             |
| Offshore QA + engineers | Accept the case<br>Owner + review window →            | Draft, review and run<br>AI proposes; engineer approves ↓   | Repair and retest<br>Fresh run → release review    |
| Platform + CI           | Provide pinned fixtures<br>Supported runner + reset → | Retain original evidence<br>Run ID + failure + next owner → | Reproduce the failure<br>Escalate missing evidence |

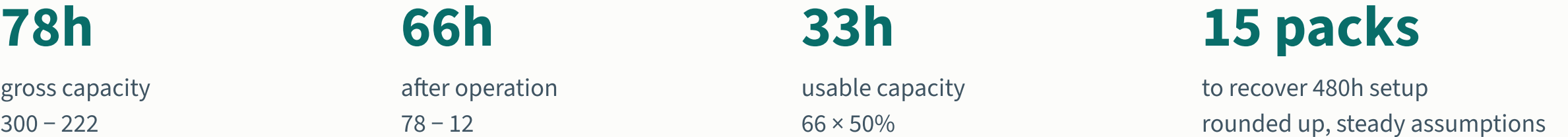
Pilot team: one QA lead, four domain testers, two automation engineers and one data/environment engineer. Product, development and platform support contribute separately.

# Effort includes review and operation

Record time at the eight workflow stages using the same boundary for baseline and assisted packs.



Purple segment = review. Total assisted effort = 222h.



Targets assume verified adoption prerequisites. Negative capacity remains possible; do not extrapolate one pack to all 75 staff or combine overlapping benefits.



# Pilot validation and stop conditions

| STEP           | PROPOSED VALIDATION                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------|
| Design         | Baseline one current pack. Compare at least two assisted packs with similar scope and matched conventional tasks. |
| Record         | Capture task complexity, reviewer time, corrections, failures, waits and environment incidents.                   |
| Stop or repair | Pause expansion for an escaped money-movement defect, unreliable fixtures or tests that pass the injected fault.  |
| Expand         | Require trustworthy mandatory coverage, positive net capacity and a successful second-squad reuse trial.          |

Small pilot samples support a delivery decision with uncertainty. They do not establish a general causal productivity rate.

# Brainstorm questions: API architecture

1. “Can we deploy a known build, reset the data and replay a provider timeout today?”

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2. “Where do payment outcomes become observable: API, journal, callback or batch file?”

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3. “Which framework already has maintainers and reliable CI execution?”

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4. “Who owns reusable fixtures and who approves the expected financial behavior?”

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Review the workflow map and a real failed release together. The first implementation backlog should follow those answers.

# Platform prerequisites for the pilot

SCOPE FOLLOWS PROVEN CAPABILITY

**Inputs + people + evidence + approved AI access + a measured baseline**

DRAFTING

**Reviewed scenarios**

Approved rules and observable outcomes.



**Domain QA accepts the scenario matrix.**

Assess test design →

DIAGNOSIS ONLY

**Evidence-linked triage**

Retained failures, known dispositions and usable diagnostics.



**A reviewer checks the diagnosis; no retest runs.**

Assess diagnosis →

EXECUTION

**Reviewed test changes**

Add repeatable environments, CI, fixtures, provider control and reliable runners.



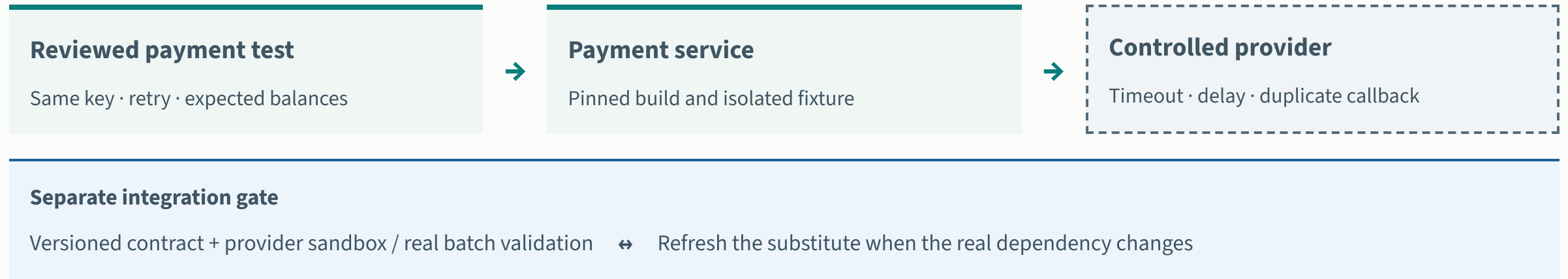
**Independent assertions challenge the candidate.**

Assess automation →

Unknown evidence remains a gap. Platform owners, reviewers and supplier leads must accept named actions and review dates before adoption.

# Service virtualization and real integration

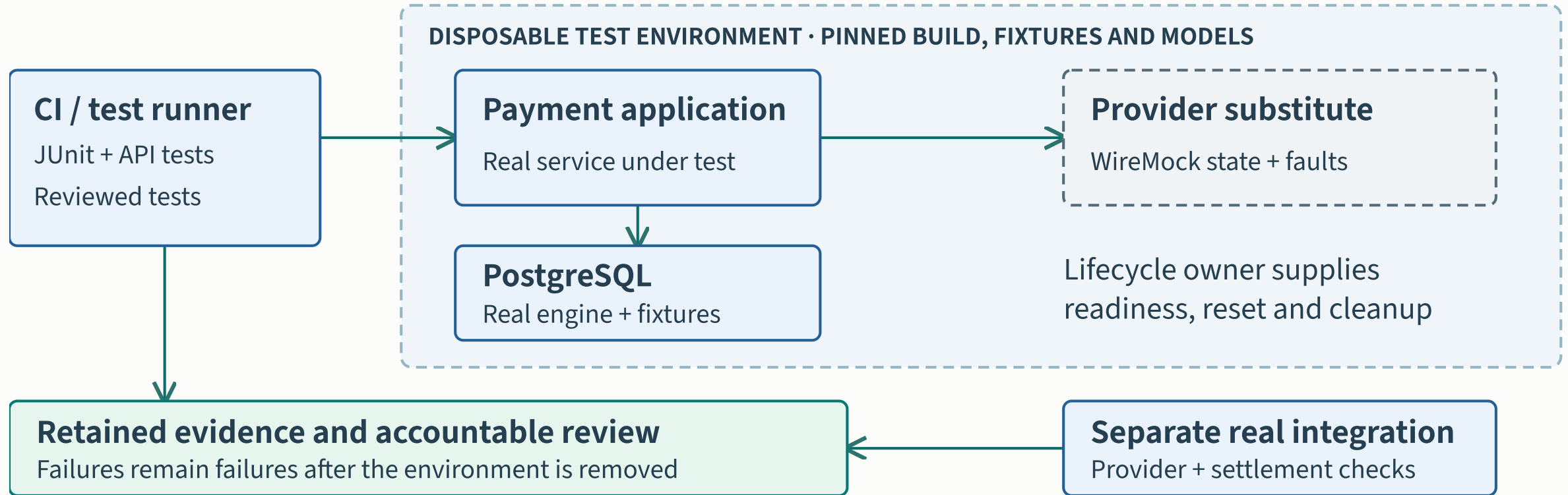
Control the payment provider for repeatable tests; keep a separate real-integration gate.



Proposed payment test architecture. A passing substitute is evidence about the modeled behavior, not proof of compatibility with every real dependency. [Dependency strategy and sources](#) →

WireMock is an HTTP example. Contract fidelity, callback behavior and batch access need explicit owners; passing a stub does not prove end-to-end correctness.

# FinTech Client: test environment



The CI runner checks the real payment service and database against reviewed provider models. Retain separate provider and settlement evidence.

# Technology roles in the payment pilot

## RUN

### Containerization

Package and run real software with repeatable dependencies.

#### EXAMPLE

A PostgreSQL instance created for one test run.

## SIMULATE

### Service virtualization

Replace selected dependency behavior with a controlled model.

#### EXAMPLE

A WireMock provider with an intentional timeout.

## CHECK

### Contract testing

Check whether consumer and provider interactions agree.

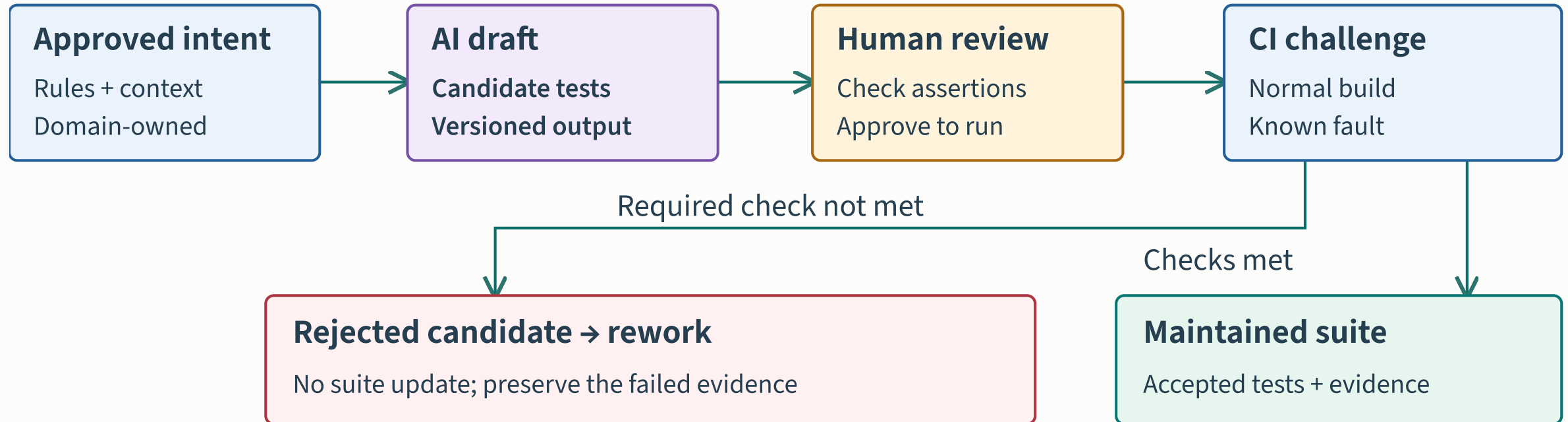
#### EXAMPLE

A provider change fails a reviewed Pact contract.

These capabilities complement each other. They do not establish complete business correctness or replace all real integration checks. [Technology choices and limitations](#)

Choose Testcontainers, Compose or an existing supported environment for the test scope. Mobile devices and legacy settlement retain their own integration requirements.

# A candidate test must earn acceptance



PAY-142 must catch an incorrect transfer, journal or balance. The guided example rejects the candidate and leaves the maintained suite unchanged.

# A fair comparison for the payments pilot

**Shared conditions** · Comparable tasks, same framework, environment and acceptance rules

## Conventional work

Engineer drafts

↓ Reviewer checks

↓ Tests challenge behavior

↓ Retain results and effort

## AI-assisted work

AI + engineer draft

↓ Reviewer checks

↓ Tests challenge behavior

↓ Retain results and effort

**Compare accepted outcomes** · Include unsuccessful attempts, review, rework and operating costs

Proposed evaluation design informed by [ANZ's study limitations](#) and [METR's selection findings](#). Separate modernization gains from incremental AI effects.

Use comparable tasks inside the proposed pilot. Include failures and review effort; retain the existing mandatory payment checks.



# The runtime from setup to cleanup

Proposed execution lifecycle. Numbered steps show order, not elapsed time.

**1 Create and check**

Pin the build, tests and configuration. Provision supported resources and wait for health checks.

**2 Seed and configure**

Create isolated synthetic accounts. Load the owned, versioned virtual-provider behavior.

**3 Execute and challenge**

Run required tests. The deliberate duplicate-posting variant must fail its independent assertion.

**4 Retain and review**

Keep original assertions, setup diagnostics and run IDs outside disposable resources. Review omissions.

**5 Clean up on every path**

Attempt teardown after success or failure. Escalate failed cleanup to the named platform owner.

**6 Reconcile integration**

Check the real provider and settlement separately. Contract drift triggers model review and rerun.

Evidence survives cleanup. A missing or failed required check holds the release.

Repeatable execution is a platform capability. Retain failure evidence and attempt cleanup on every path; real integration remains separate.

# Connect the tools behind the platform

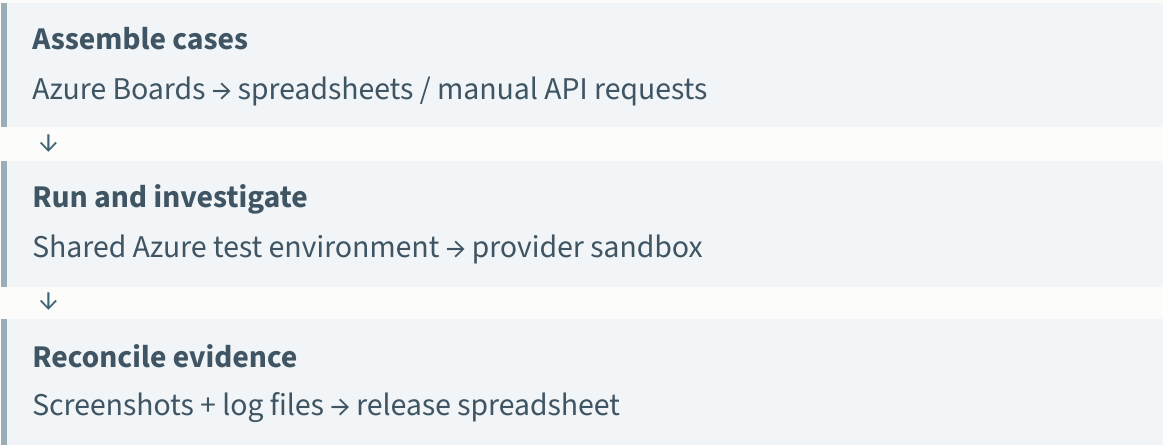


Proposed integration. Validate access; fund context adapters and joined evidence.

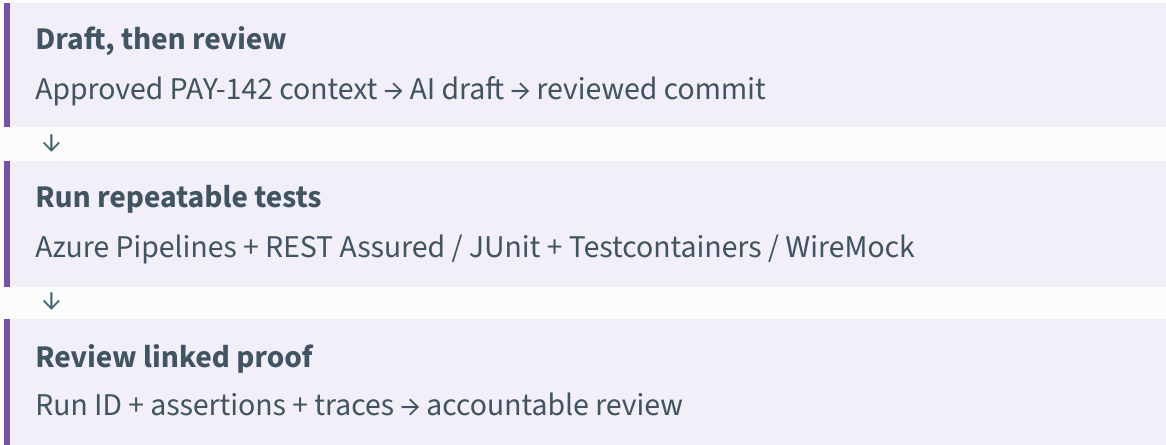
# Before / after: the API test architecture

One retry → one transfer → independently checked balances

## Before · manual API QA



## After · target pilot workflow



Requires CI, isolated data, validated provider stubs and reviewers. Keep real integration checks.

Pilot target. AI drafts; people approve intent and changes; assertions verify the payment outcome.